Inheriting the Bloom: Biometric Abstraction and Co-creation in Cross-Reality Art

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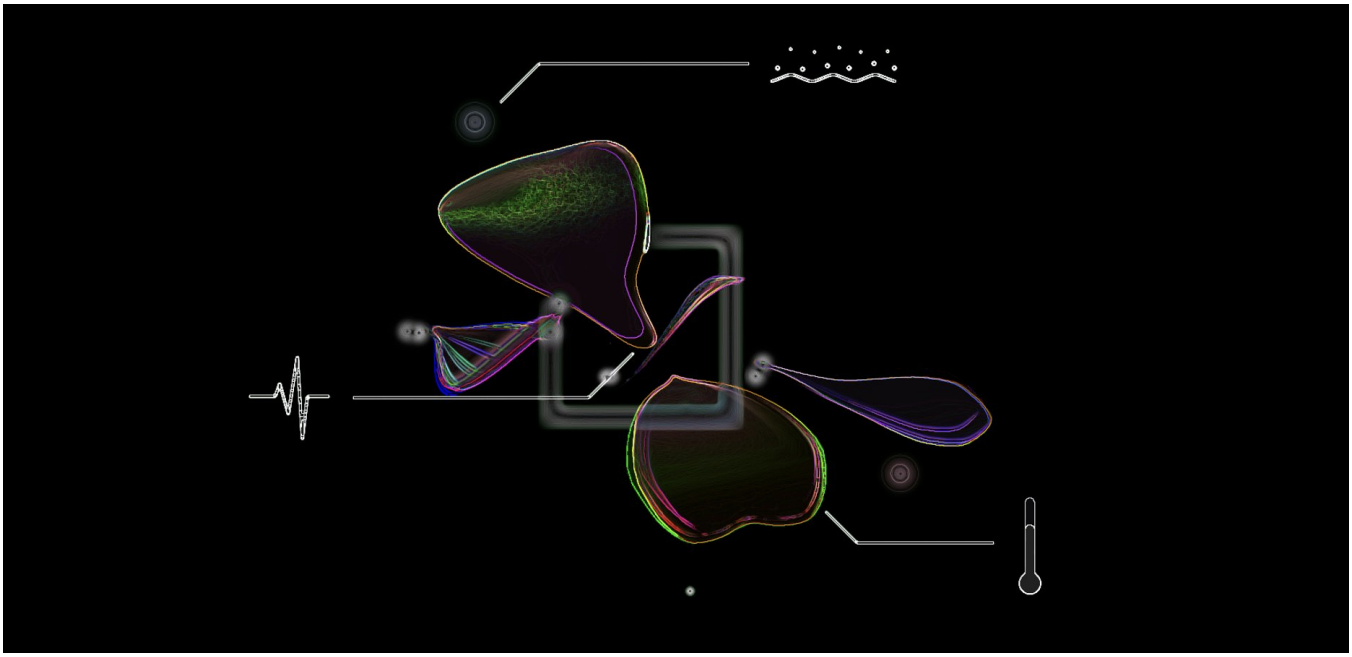


Figure 1: Participant biosignals rendered as generative flora, shaped by heart rate, skin conductance, and temperature.

Abstract

Inheriting the Bloom is a cross-reality installation that explores the creative complexities of translating bodily signals into shared aesthetic artifacts. Participants enter a quiet room containing a single pot, approach it through a watering gesture, and don an XR headset that shifts between mixed reality and virtual reality. Their biosignals, including heart rate, skin conductance, and temperature, shape a digital bloom through deliberately ambiguous mappings that resist direct physiological interpretation. The bloom joins a collective virtual tree before collapsing into a stylized seed whose abstracted traces become the next participant's inherited starting condition. Co-creation unfolds through sequential inheritance as each visitor receives traces left by others and passes them forward in altered form. Anchored in touch, the pot connects virtual transformation to material presence, while the virtual archive manifests how physiological traces inhabit and shape a shared world over time.

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CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality**;
• **Applied computing** → **Media arts**.

Keywords

Experimental Visualization, Bio-signal, Interactive Art

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1 Introduction

Biometric and physiological signals are semiotically dense. Vital sign monitors situate hospital patients along axes of medical normalcy, while security sensors test bodily traces against opaque systems of surveillance. Polygraph tests, though debunked [Iacono and Lykken 1997], continue to carry cultural authority, and somatic data increasingly replaces passwords and PINs. Across these contexts, biometrics are burdened with proof: that we are who we claim to be, that our bodies function as expected, and that our internal states can be made externally legible.

Yet biometric meaning rarely resides in the signal itself, but in how it is mediated and made perceptible. It is not the heart stopping, but the monitor's flatline tone that signifies death on screen; not the body's state, but the oscillating needle of a lie detector or the pulsing waveform of a fitness display that communicates guilt, danger, or effort. Across popular media and technical interfaces alike, biometric meaning is produced through stylized representations that stand in for the body.

This raises a broader question for artistic practice: if biometric meaning is already produced through mediation and representation, what happens when these processes are made explicit? Rather than faithfully visualizing physiological states, how might biosignals be transformed, shared, and preserved beyond the moment of measurement? What new relationships become possible when biometric data is treated not as a record of the body, but as material for collective and evolving experiences?

Inheriting the Bloom explores these questions through a cross-reality installation in which people water a physical pot, wear an XR headset, and generate a digital bloom on a virtual tree through their biosignals. The work maps biosignals through indirect and partially randomized relationships that resist direct interpretation, treating ambiguity as a design value and the body as a contributor to a shared ecology. At the end of each session, the bloom collapses into a seed returned to the pot, proposing alternative futures for biometric interaction grounded in contribution. Participation unfolds sequentially, with each participant contributing to the conditions experienced by the next.

2 Background: Translation, Transformation and Co-creation of Biometrics

Biometric data, including heart rate, body temperature, and neural activity, is increasingly used in artistic practices exploring the relationship between bodies, environments, and systems of meaning [Sparrow et al. 2024]. Most biosignal artworks rely on direct, legible mappings and visualization to illustrate liveness and co-presence. Lozano-Hemmer's *Pulse Room* [Lozano-Hemmer 2006] and Franco's *Heart of the City* [Franco 2015] translate heartbeats into rhythmic light installation and spatial sculpture, while Kim's *Heart Space* [Kim 2024] turns individual heartbeats into unique chromatic waveforms in real time as distinct visual identities. Related paradigms similarly repurpose electroencephalogram (EEG) or eye-tracking data for real-time performance [Han and Surve 2019; Hughes and Jorda 2021; Jiao 2024]. Across these works, the body is made perceptible through clear mappings between signal and form.

A few examples shift away from direct representation toward interpretive and more abstract transformation. Lozano-Hemmer's later installation *Thermal Drift Density Map* [Lozano-Hemmer 2022] transforms visitors' body heat into slow-moving particle clouds, rendering warmth as a dispersive emission that blurs the boundary between body and environment. Putnam's *Emergent* [Putnam 2023] feeds heart rate and movement data into pseudorandom generative animations and sound compositions, producing geometric patterns designed to abstract the body states.

A growing body of work further extends this shift by bringing multiple participants' biosignals into shared co-creation. *InterPulse* [Chan 2020] invites participants to collaboratively visualise their pulses as combined particle systems, supporting social connection. *My Virtual Dream* [Kovacevic et al. 2015] places 20 participants in a geodesic dome, using collective EEG states of relaxation and concentration to drive visual projections in real time. Wang et al.'s *SynCocreate* [Wang and Feng 2024] similarly maps EEG synchrony between pairs of users into colors and strokes on a shared 3D canvas, emphasizing relational emergence over individual measurement.

These systems share two characteristics: participation is typically bounded within a single session, and biosignal mappings operate in real time without persistence beyond the interaction. While effective for situated co-creation, they generally do not carry traces forward across encounters. The present work extends this direction by exploring how biosignal traces might persist across sessions as inherited conditions that shape future interactions, spanning virtual and physical space.

3 Conceptual Framework

3.1 Productive Ambiguity as Resistance to Surveillance

While many biometric systems are designed around legibility to make biosignals readable and comparable [Halbig and Latoschik 2021; Sparrow et al. 2024], we propose ambiguity as an alternative design value. The signals are assigned to visual parameters through noise, constrained ranges, and partially randomized relationships. No direct reading of the biosignal is possible: the bloom shifts and changes, but never declares what it means.

3.2 Biosignal as Collective Traces

In contrast to biometric paradigms centered on data extraction, we frame biosignals as traces within an environment sustained through cross-session participation. Individual traces accumulate within a collective tree, persisting not as retrievable records but as baseline conditions for subsequent visitors. This framework replaces models of individual ownership with a system of custodianship [Sanders and Stappers 2008], wherein participants receive inherited traces and pass them forward transformed.

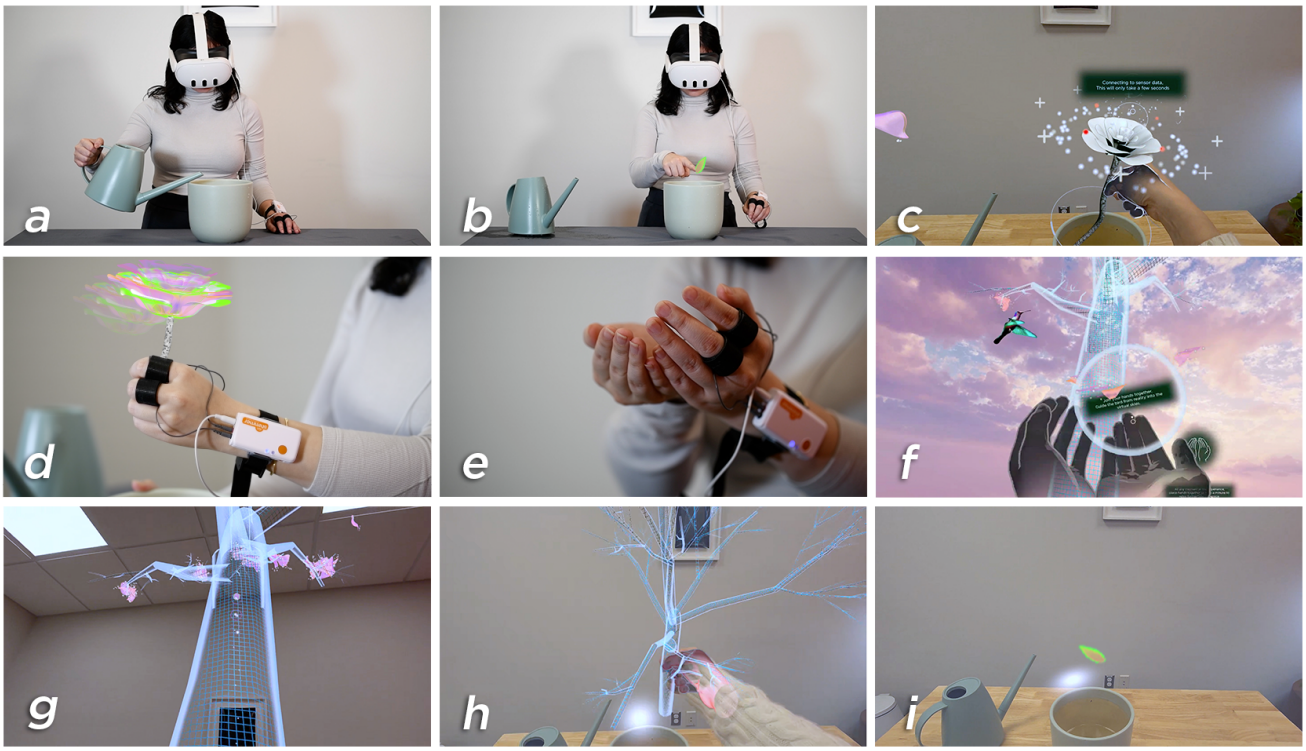


Figure 2: Experiential and inheritance flow across physical and XR space. (a) Watering the shared pot begins the encounter; (b) an inherited seed establishes the starting state; (c–d) biosignals shape a new bloom; (e–f) a hand gesture shifts the experience from MR to VR; (g) the bloom joins prior contributions in the shared virtual ecology; (h) the contribution is completed at the tree; (i) the bloom returns as a seed whose derived parameters condition the next encounter.

3.3 Continuity Across Virtual and Physical Action

Our design also involves physical action that bridges virtual and material experience. Gestures such as watering the pot, donning the headset, and placing the seed back into the pot function as embodied expressions of the experience. These actions connect bodily responsibility to the virtual logic of the art, performing a cycle of emergence, relinquishment, and continuation. Requiring participants to physically return this interactive residue ensures that cross-reality continuity is felt as an ongoing process rather than merely represented symbolically.

4 The Artwork - Inheriting the Bloom

4.1 Physical Setup and Threshold

The experience begins in a quiet room with a single physical pot placed at its center. Participants approach the pot and water it to begin the XR experience. This gesture establishes a material threshold between everyday space and the translated environment that follows, grounding entry in touch rather than an interface command.

Watering functions as a preparatory act that slows the participant and situates their body in relation to the work. It frames participation as care and responsibility, introducing the cyclical logic that underpins the experience: emergence, reduction, and return. The pot persists across sessions as the site where traces are received and relinquished, anchoring custodial continuity in an object that can be held, tended, and passed on.

4.2 Experiential Flow

Inheriting the Bloom unfolds as a sequential experience (Figure 2) in which each participant inherits traces from a prior visitor. The session begins at the pot, where an inherited seed establishes the starting state. In MR, the participant's biosignals shape a digital bloom through real-time changes in color, motion, scale, and texture. The bloom then joins the shared virtual tree alongside prior contributions. At the end of the session, it is reduced to a stylized seed whose derived visual parameters are stored locally and returned to the pot for the next participant, completing the cycle without preserving raw biometric data or revealing the previous visitor's identity or intent.

4.3 XR Experience

4.3.1 The Visuals. The visual language shapes biosignals as living, expressive material (Figure 3). We created shaders for separate parts of the flora, allowing attributes such as color, hue intensity, and texture noise to respond in real time to the physiological stream. The petals are rendered with layered transparency to maintain

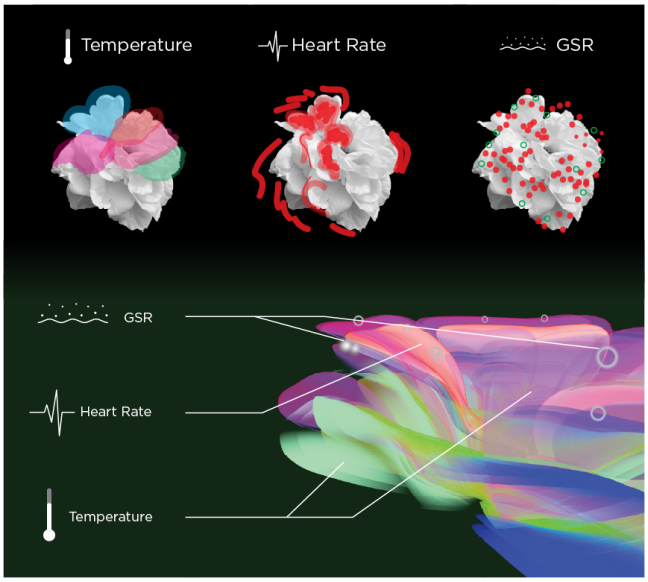


Figure 3: Real-time shader art using layered Fresnel and noise-based parameters, driven by sensor input.

a sense of permeability between virtual form and physical space. A subtle fresnel effect further accentuates the petals' soft edges, enhancing depth and highlighting shifts in the bloom as signals fluctuate. As the signals oscillate over time, mapped properties produce continuous micro-movements across the petals. Ambiguity is also embedded at the implementation level. Each shader parameter responds to a composite value derived from multiple signals combined through weighted noise functions, preventing any single physiological input from producing a directly legible visual change.

4.3.2 Simultaneous Spaces. The experience invites participants to inhabit simultaneous physical and virtual spaces through a reversible transition between MR and VR. This shift functions both as an interaction mechanic and an artistic gesture, staging the passage of physiological traces from bodily sensing into immersive form. In MR mode, passthrough maintains awareness of the room, while VR mode introduces a fully virtual environment. Participants toggle between modes through a simple hand gesture, and both the participant and the bloom persist across transitions, reinforcing continuity of presence across realities.

4.3.3 The Sound and Music. The experience features an integrated soundscape in which ambience, music, and effects respond to the participant's body. Audio layers probabilistically fade in and out, while the biosignal stream modulates the balance between naturalistic textures (e.g., forest ambience) and musical elements (e.g., piano), alongside global processing such as reverberation and spectral filtering. Certain sounds are hard-coded to events such as the MR to VR transition, emphasizing shifts between perceptual states. These cues occur only in-headset, while a continuous ambient layer is broadcast through physical speakers, grounding the experience in the room's acoustics. Within this structure, each visitor shapes a personal sonic layer, while the broader soundscape persists as a shared environment passed from one participant to the next.

4.3.4 Technical Implementation and Persistence. The XR system was developed in Unity 2022.3.60 and runs on a Meta Quest 3. Biosignals are captured via a Shimmer3 GSR+ sensor and streamed into Unity through a custom integration[1]. Adaptive audio was implemented in FMOD[2]. Raw physiological signals are mapped in real time to bloom attributes, including petal shader parameters and spatial position, but are not retained. At the end of each session, these derived visual parameters are serialized to a compact JSON file in Unity's persistent data path. The stored state preserves completed blooms within the collective tree (Figure 4) and shapes the seed inherited by the next participant, maintaining continuity through aesthetic traces rather than raw biometric data [Wei et al. 2025].

5 Exhibition Reflections

An earlier iteration of the installation, titled *The Arborist* [Wei et al. 2026], was exhibited at the IEEE VR XR Art Gallery (Daegu, Korea, March 2026), where more than 60 visitors encountered the work over three days. Attendees included XR researchers, psychologists, engineers, artists, and members of the public. Over 90% completed the full session, with an average engagement time of approximately two minutes. The reflections below are practice-based observations drawn from informal conversations and on-site encounters rather than a formal audience study.

5.1 Biosignal Awareness and Meaningful Illegibility

Visitors frequently tested the system through posture, breathing, and movement in an attempt to understand how their bodies shaped the bloom. One asked which physiological signal controlled each visual change; another ran on the spot to observe effects; a third wondered whether holding their breath would keep the virtual bird in their hands longer. A recurring question was: "What is the flower saying about my body?"

Some visitors who could not fully decode the mapping continued experimenting, while others reached brief moments of recognition—"Cool, so that is my flower"—and continued exploring. These encounters suggest that ambiguity may sustain curiosity, but they also surfaced uncertainty about agency. One visitor remarked, "I wish I could see more dramatic changes," indicating that abstraction may become less generative when bodily effects are too subtle to perceive. This tension aligns with ambiguity as a design resource [Gaver et al. 2003], while also suggesting that productive illegibility depends on preserving enough perceptual contingency for participants to sense that their bodies matter.

5.2 Collective Biosignals as Ecological Contribution

Visitors also recognised the collective dimension of the shared tree. One asked, "Are those the other people's flowers?" Others expressed interest in returning to observe how the system had changed, or

[1]https://github.com/shuwithu/Unity-Shimmer-DataStreaming
[2]https://www.fmod.com/

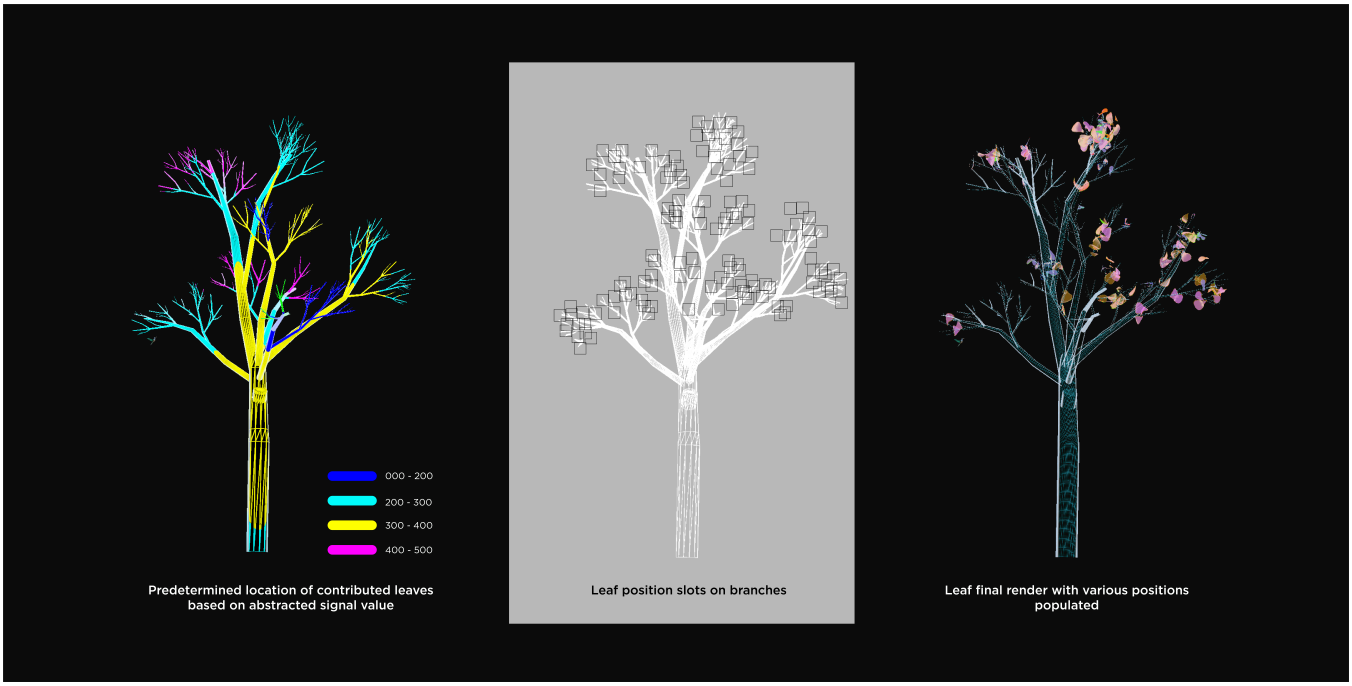


Figure 4: Abstracted signal values shape the spatial distribution of blooms within the collective tree.

speculated about what the tree might look like for a visitor arriving on the final day of the exhibition.

Interpretations varied. One visitor imagined the work in a co-working or wellness environment where physiological traces accumulate within a shared environment; another compared the tree to a journal in which each bloom marks a moment to be revisited, seeing its potential as an "artistic mental health app". At the same time, the relationship between an individual bloom, the shared tree, and the inherited seed was not always clear. This points to a design challenge for future iterations: inherited traces may need to become more perceptible without becoming fully legible or individually attributable.

6 Discussion and Conclusion

We treat biometric sensing as an artistic site of translation, proposing ambiguity as a productive condition and biosignals as contributors to a virtual ecology. Unlike prior biometric artworks that operate within single-session synchronous co-creation, *Inheriting the Bloom* structures participation sequentially, where physiological traces become conditions for future encounters. Exhibition observations suggest that unclear mappings can prompt experimentation and interpretation, but also reveal a boundary between productive ambiguity and arbitrariness. Meaningful illegibility depends not on eliminating causal connection, but on allowing participants to perceive that their bodies matter without fully decoding how.

Custodianship provides the mechanism connecting ambiguity and ecological contribution. Each visitor inherits traces left by a predecessor and contributes conditions that shape those who follow, making continuity a matter of shared responsibility rather than individual control. The archive is therefore not a stable store of biometric records, but a changing environment sustained through ongoing transformation.

The installation also surfaces tensions between abstraction, differentiation, and fidelity. Translating physiological signals into visual form involves compression and loss, raising questions about how much variation can be abstracted before individual traces become indistinguishable. Rather than treating this solely as a limitation, we make this boundary visible as part of the artistic inquiry. Future iterations could explore stronger perceptual differentiation, more perceptible inheritance cues, or adjusted participant guidance, while preserving the refusal of direct biometric readback.

Ultimately, *Inheriting the Bloom* reframes the biometric interface as a medium for relational translation. It proposes a cross-reality model in which physiological traces persist through contribution, inheritance, and transformation rather than measurement and static storage. Rather than asking what the body reveals, this paradigm asks how physiological traces might participate in shaping collective worlds across people, places, and time.

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